A Metapopulation Model for the Effect of Invasive Lionfish on Coral Reefs

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We pledge our honor that we have not violated the honor code during this assignment.

1 Introduction

Our project seeks to understand and model how human intervention strategies can be used to support healthy coral systems. We primarily focus on determining how harvesting invasive lionfish populations at different fishing efforts variably affects the maintenance of healthy coral populations.

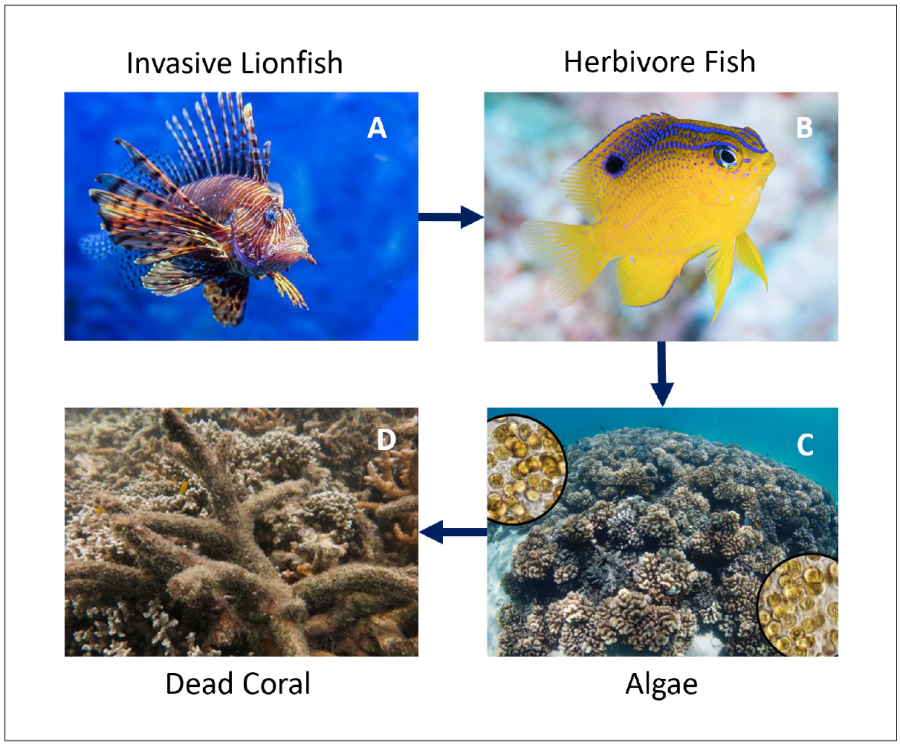

Figure 1: Relationship between the invasive lionfish population and coral reefs. Lionfish (A) are known predators of herbivore fish (B) that consume algae (C) on coral reefs. Overgrowth of algae on coral creates a hypoxic environment that eventually leads to the death of corals (D). Overpopulation of lionfish in this ecosystem leads to the loss of coral reefs.

Supporting more species per unit area than any other marine environment, coral reefs are among the most valuable ecosystems in the world [1]. Healthy coral reefs are important for sustaining commercial and subsistence fisheries through providing thousands of species of fish for harvesting [6]. Moreover, by serving as shoreline buffers to significant amounts of energy from storms, wind, and waves, healthy corals ensure that people and habitats are protected from erosion and property damage [1].

Despite their abundant contributions to human health and happiness, coral reefs are threatened by various anthropogenic activities. Pollution, habitat destruction, and the introduction of invasive species can force reefs out of stable coral-dominated states [7]. This project will chiefly examine the effects of lionfish invading coral habitats.

As an invasive species, lionfish have few predators and can propagate rapidly in a given region [8]. As generalist feeders, lionfish primarily wreak havoc on coral ecosystems by consuming and thereby significantly reducing herbivorous populations [8]. Indeed, researchers have estimated that a single invasive lionfish can decrease recruitment of native fish by 79 percent [3] in as soon as 5 weeks. In high productivity marine environments such as open-ocean-facing reefs, reductions in herbivorous feeders enable algal blooms, which interfere with coral recruitment, suppress coral growth and fecundity, and can directly kill certain species of coral via oxygen depletion [7]. This relationship can be seen in Figure 1. As effects traverse trophic cascades, the loss of corals is strongly associated with a decline in reef biodiversity [7]. Ensuring the survival of healthy corals is necessary for conserving the great biodiversity of our reefs.

Such damages have direct effects on communities located near coastal areas, as well as on industries that rely on tourism and fisheries. Mitigating the effects of invasive lionfish is therefore a high priority for many groups. Although efforts to reduce lionfish populations have often been spearheaded by individual spear fishers, researchers are now exploring how market-based strategies must be considered to efficiently control growing lionfish populations [4, 8].

2 Background

2.1 Metapopulation Models

Metapopulation or compartmental models are a powerful tool for analyzing epidemics. These discrete-transition models can provide useful insights about the spread of disease. For example, Susceptible-Infected-Recovered (SIR) models describe the disease driven transitions between susceptible, infected, or recovered classes for individuals in a population. These models consist of Ordinary Differential Equations (ODEs) which represent the dynamics of specified subpopulations. Specifically, an SIR model is composed of three ODEs to represent the dynamics of the susceptible class, infected class, and recovered class:

$$\begin{aligned}\frac{dS}{dt} &= -\beta SI,\\ \frac{dI}{dt} &= \beta SI - \gamma I,\\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

Here, the entire population is the sum of individuals in the susceptible (S), infected (I), and recovered (R) classes (e.g. $S+I+R=1$). Other parameters represent specific qualities of a given disease. The parameter β represents the rate of infection, and γ represents the recovery rate of an individual. The SIR model can be expanded to account for vaccination strategies, birth-rate (continually introducing new individuals to the susceptible class), and death-rate (individuals leaving the population due to natural or disease driven death).

Additionally, we can consider the Susceptible-Infected-Susceptible (SIS) Model in which there are only two classes: susceptible and infected. Here, individuals transition between the two classes. Generally, an SIS model has two ODEs that are defined by the following equations:

$$\frac{dS}{dt} = -\beta SI + \gamma I$$
$$\frac{dI}{dt} = \beta SI - \gamma I$$

Here, the entire population is the sum of individuals in the susceptible (S) and infected (I) classes (e.g. $S + I = 1$). Similar to the SIR model, β represents the rate of infection, and γ represents the recovery rate of an individual.

While compartmental models such as SIR and SIS are useful for modeling epidemics, they can also be applied to a broader class of problems in ecological systems, see for example [9], where they use these models to examine the transitions between vegetated, fertile and deserted soil. For this project, we couple a compartmental model that represents the relationship between coral colonized by algae, and a predator-prey model between a herbivore fish population, and a lionfish population. Ultimately, we seek to understand how harvesting lionfish effects the health of coral reefs. To do this, we include a harvesting term in the differential equation that describes the population dynamics for lionfish. This term will be modified to examine how different harvesting rates affects the long term health of corals in terms of colonizaton rates.

2.2 Optimal Lionfish Yield

Models indicate that commercially harvesting lionfish at rates above the maximum sustainable yield (MSY) may substantially reduce population sizes [4, 5]. Researchers propose that fisheries should harvest lionfish at the Optimum Lionfish Yield (OLY), a target that exceeds MSY [4] (see Figure 2). The Maximum Sustainable Yield (MSY) describes the long-run harvest that does not push the species population to extinction [5]. To ensure the sustainable survival of a species, fishers typically must harvest at rates below the MSY. However, because invasive lionfish are extraordinarily robust to typical fishing pressures, the OLY provides a rate of harvest above the MSY at which lionfish can be sustainably harvested while also reducing their impact on ecosystems [4]. The OLY-MSY comparison can be seen in Figure 2. Indeed, their model suggests that fishing at the OLY would reduce lionfish populations by an additional 42 percent relative to fishing at the MSY [4]. Despite these projected successes, fisheries presently have little incentive to fish at levels beyond the MSY, as increased fishing efforts would be required for the same or a potentially lower yield over time [4]. Through our project, we seek to demonstrate the necessity of harvesting lionfish at OLY levels or above to preserve the health of coral reefs.

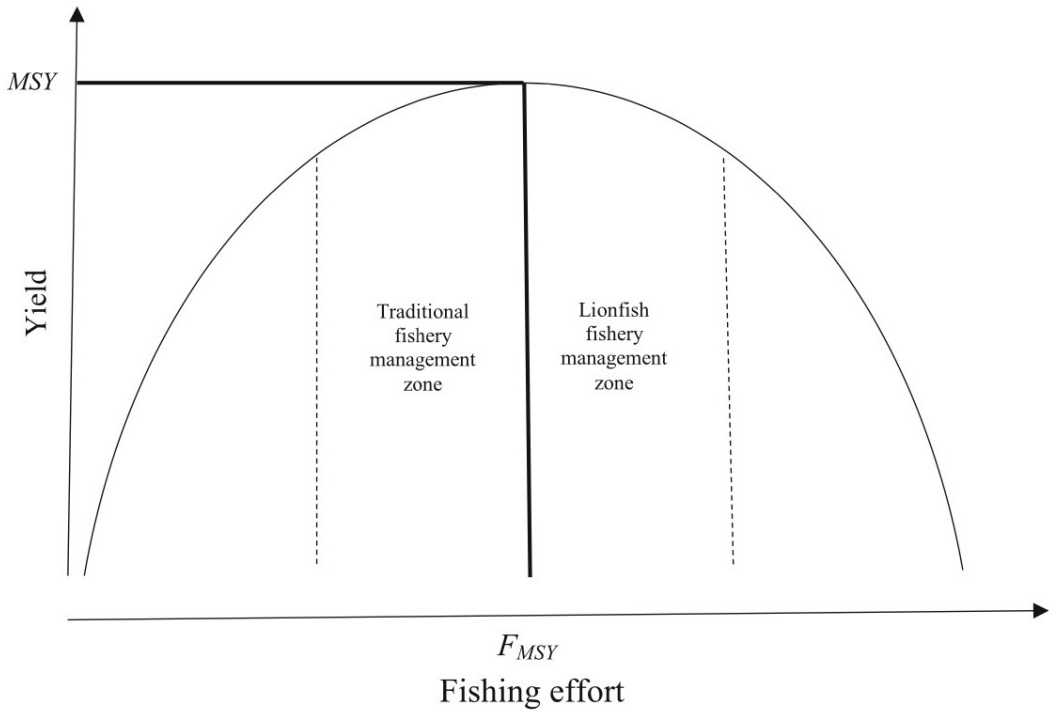

Figure 2: A generalized sustainable yield curve depicting the approximate zones of traditional fishery management targets compared to proposed lionfish fishery management targets. Figure from [4].

3 Our Metapopulation Model

Consider a metapopulation model representing the dynamics of coral reef, herbivore, and lionfish populations. First, suppose coral can be either colonized or uncolonized by algae (C). We model the colonized coral population with the following differential equation:

$$\frac{dC}{dt} = \beta C(1-C) - eC - \rho CH.$$

Here, coral becomes colonized by algae at a rate β, and becomes uncolonized at a rate e. H represents the herbivore population, and ρ is the rate at which herbivore fish feed on colonizing algae.

Within our system, lionfish (L) consume herbivorous fish. We model the relationship between lionfish and herbivore populations using the following predator-prey dynamics:

$$\frac{dH}{dt} = dH - \gamma LH$$
$$\frac{dL}{dt} = rL\left(1 - \frac{L}{K}\right) - xL.$$

The herbivore population (H) is modeled using exponential growth. Herbivorous fish grow at a constant rate α and are actively predated on by lionfish at a rate γ. The lionfish population follows

logistic growth where r is the lionfish growth rate and K is the carrying capacity of the system. Corals experience an increased algal colonization due to the predation of herbivorous fish. Our goal is to understand the how the colonization of corals is affected by lionfish harvesting effort. Here, we have established a tuning parameter x to represent the harvesting effort. The complete dynamical system is as follows:

$$\begin{aligned} \frac{dC}{dt} &= \beta C(1-C) - eC - \rho CH, \\ \frac{dH}{dt} &= \alpha H - \gamma LH, \\ \frac{dL}{dt} &= rL\left(1 - \frac{L}{K}\right) - xL. \end{aligned} \tag{1}$$

All parameters have been detailed in Table 1. The lionfish population is assumed to rise to 1.5 million in the absence of fishing pressures, as estimated by Bogdanoff et al [4]. The herbivore fish population, which encompasses significantly more species than lionfish, is approximated to be ten times greater than the lionfish population before predation takes effect.

Parameter	Definition	Value
C	Initial Proportion of Colonized Coral	0.0001
H	Initial Herbivore Population	15
L	Initial Lionfish Population	1.5
β	Colonization Rate of Algae	5
e	Uncolonization Rate	2.5
ρ	Rate at Which Herbivores Feed on Algae	3
γ	Predation Rate of Lionfish on Herbivores	2
α	Growth Rate of Herbivores	0.5
r	Lionfish Growth Rate	0.71
K	Carrying Capacity	3
x	Harvesting Rate of Lionfish	See Table 2

Table 1: Parameters for metapopulation model. The values in the third column are used for the simulations presented in the next section. All values are scaled by 10^{-6}.

The lionfish harvesting rate, x, is a function of fishing effort such that

$$x = \alpha \cdot \text{Effort} \tag{2}$$

where α is a constant. For our simulations, varying harvesting rates of lionfish were tested. These rates can be seen in Table 2. To simplify our calculations, we assume that $\alpha = 1$ such that x directly corresponds to fishing effort values.

In our model, we scale fishing effort to be a value between 0 and 1. Therefore, high fishing effort

is represented by 1. Further, according to material presented in Lecture 8, the lionfish harvesting effort at MSY is half the maximum fishing effort. Therefore the lionfish harvesting effort at MSY is 0.5. [4] detailed current lionfish harvesting effort to be below MSY, and can be approximated to a harvesting effort of 0.33. The OLY fishing effort must be greater than that of MSY but below the high fishing effort. This is because we need a value that will significantly reduce but not eradicate the lionfish population. After running our simulations with different values, we determined 0.7 to be a value that fit this criteria. Lastly, a low fishing effort was selected to be 0.05. These harvesting rates have been visualized in Figure 3.

Fishing Effort Scale	Value
Low	0.05
Current	0.33
Maximum Sustainable Yield (MSY)	0.5
Optimum Lionfish yield (OLY)	0.7
High	1

Table 2: Parameters for harvesting rate of lionfish. The values in the second column were interchanged for our simulations presented in the following section.

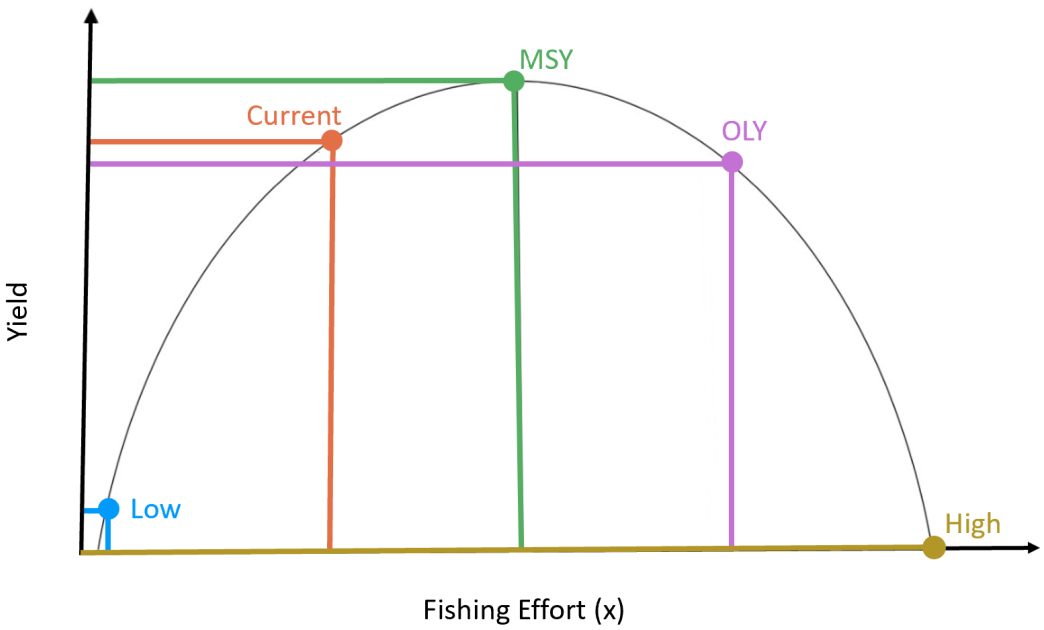

Figure 3: Long-term lionfish yields as a function of these different lionfish harvesting efforts will be tested using our model. Figure adapted from [4].

3.1 Equilibrium Points

To understand the long term effects of the interaction between these populations, we solve our dynamical model to obtain analytical solutions for their equilibrium points. The equilibrium points of this system are as follows:

$$
\begin{aligned}
&\bar{L} = 0, \\
&\bar{L} = K\left(1 - \frac{x}{r}\right) \\
&\bar{H} = 0, \\
&\bar{H} = H \textbf{ when } \bar{L} = K\left(1 - \frac{x}{r}\right) = \frac{\alpha}{\gamma} \\
&\bar{C} = 0, \\
&\bar{C} = \frac{\beta - e - \rho H}{\beta}
\end{aligned}
$$

These equilibria result in one of two possible scenarios for lionfish populations: extinction ($\bar{L} = 0$), or a stable equilibrium determined by their carrying capacity, growth rate, and fishing effort ($\bar{L} = K\left(1 - \frac{x}{r}\right)$).

When lionfish go extinct, there are 2 possible equilibria for herbivorous fish: extinction before the lionfish population goes extinct, or they survive longer than lionfish and proceed to grow exponentially. In either case, colonized coral can either survive at some stable equilibrium or go extinct. This scenario is given by:

$$
\begin{aligned}
\bar{L} &= 0, \\
\bar{H} &= 0, \\
\frac{dH}{dt} &= \alpha H \\
\bar{C} &= 0, \\
\bar{C} &= \frac{\beta - e - \rho H}{\beta}
\end{aligned}
$$

Alternatively, if the lionfish population survives a given harvesting effort and reaches a stable non-zero equilibrium, then in most cases herbivore populations will go extinct. For a very specific combination of parameter values, $K\left(1 - \frac{x}{r}\right) = \frac{\alpha}{\gamma}$, the herbivore population reaches an arbitrary stable equilibrium. However, since the actual values of these parameters are unlikely to satisfy this specific relationship, we can ignore this scenario in our simulations. Once again, colonized coral can reach either of the two same equilibria as in the case where lionfish go extinct. This

scenario is given by:

$$\bar{L} = 0,$$
$$\bar{H} = 0,$$
$$\bar{H} = H \textbf{ when } \bar{L} = K\left(1 - \frac{x}{r}\right) = \frac{\alpha}{\gamma}$$
$$\bar{C} = 0,$$
$$\bar{C} = \frac{\beta - e - \rho H}{\beta}$$

4 Simulation Results

We ran simulations of our metapopulation model using the values in Tables 1 and 2. In Figure 4 we see how the lionfish and herbivore fish populations evolve over time for different harvesting efforts (x). When lionfish harvesting is low, current, or at MSY, the herbivore fish population will eventually go extinct. This matches our expectation given that invasive lionfish can decimate a local herbivore fish population by 79% in less than 5 weeks [4]. When lionfish harvesting efforts are at the OLY or high, the lionfish population continually declines and the herbivore fish population declines at first but eventually recovers. Notably, at the highest fishing rate, the lionfish population eventually goes to extinction, while at fishing at the OLY enables the lionfish population to persist at a low population size. Fishing at the OLY therefore would significantly reduce the lionfish population to a size that would enable healthy corals to persist but still enable fisheries to have a sustained supply of lionfish.

In Figure 5 we see how the proportion of colonized corals evolves over time for different lionfish harvesting efforts (x). For low, current, and MSY lionfish harvesting rates, the coral reef populations eventually become fully colonized. When lionfish harvesting is at the OLY, many corals become colonized, but eventually are able to become uncolonized and recover. When harvesting lionfish at a high rate, the corals never become colonized because all of the lionfish are harvested very quickly, and the lionfish population is unable to recover.

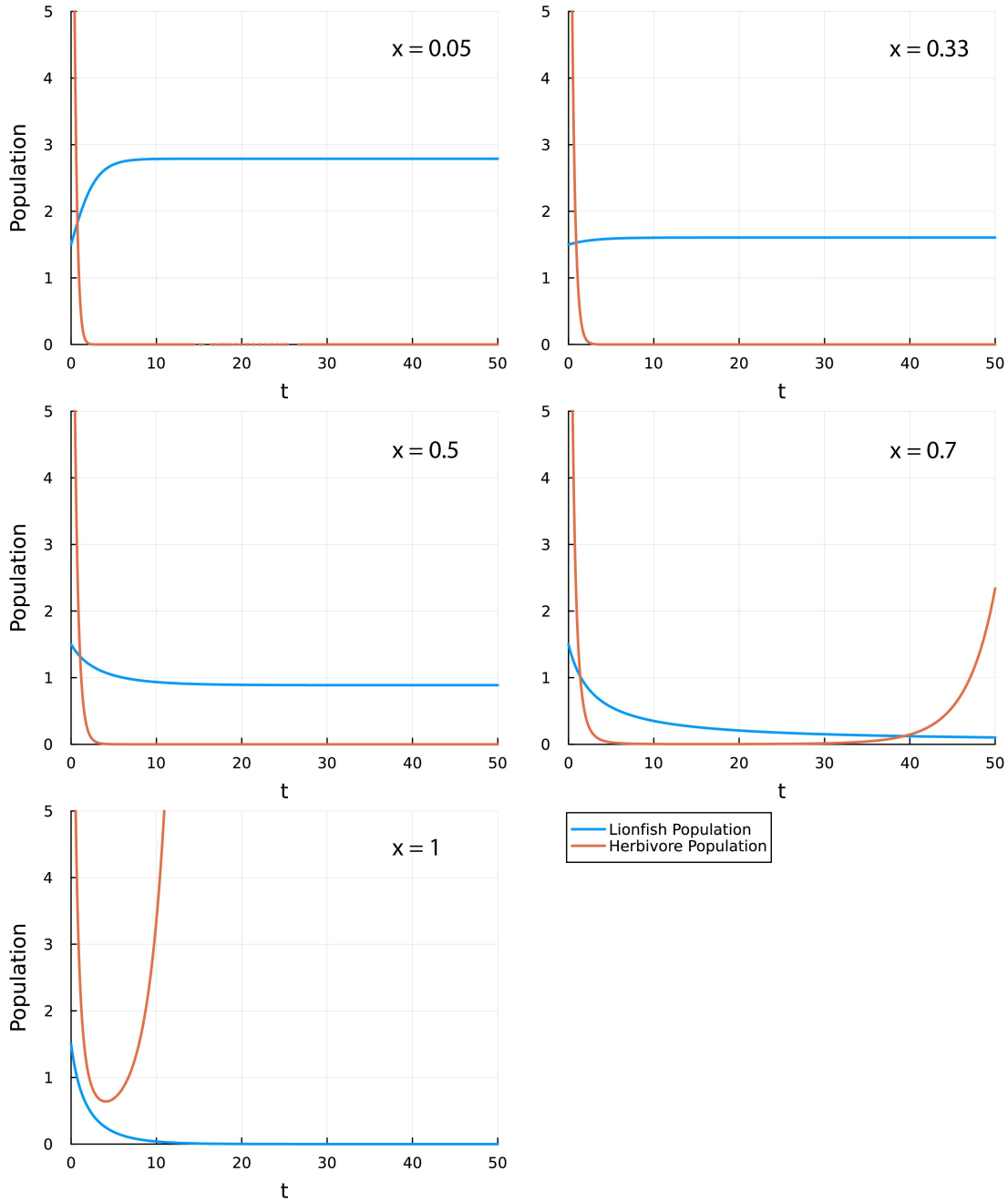

Figure 4: Lionfish and herbivore populations over time after varying x, the harvesting effort. Notice that for $x = 0.7$ and $x = 1$ (i.e. higher harvesting efforts), the lionfish population decreases ans the herbivore populations explode. On the contrary, for smaller values of x, the herbivore population decreases quickly and reaches the value 0.

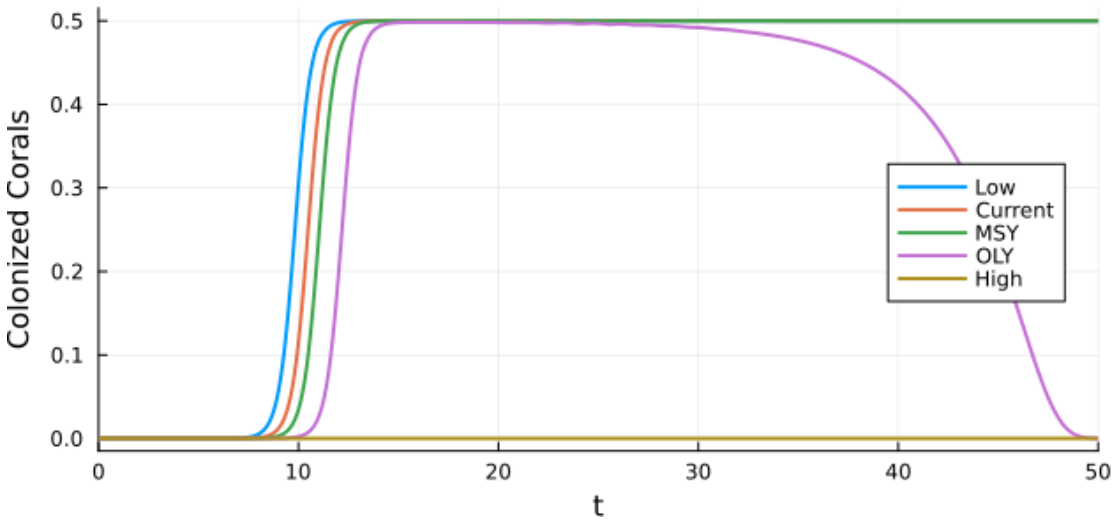

Figure 5: When the harvesting efforts are at the Optimum Lionfish Yield (OLY) or higher, the proportion of colonized corals reaches 0 **at the steady state. Otherwise, i.e., while fishing below the OLY, the proportion of colonized corals increases and reaches a non-zero value.**

5 Discussion

Our model successfully demonstrates the importance of harvesting lionfish at or above the OLY for the purpose of maintaining coral reef health. While our model provides promising insights, there are various methods that can be used to enhanced our model.

For the model defined by Equation (1), we assume that the growth rate of the lionfish population does not depend on the herbivore population. We made this assumption because lionfish are known to feed on many species [2], and therefore, even in the absence of herbivore fish, the lionfish population will continue to grow. Additionally, for our model we assume linear growth for herbivore fish. For greater lionfish harvesting efforts, this assumption leads to an unbounded growth of the herbivore fish population. We understand that this assumption is not an accurate representation of reality. To explore this assumption, we show simulations for the same parameters and initial conditions while using logistic growth of herbivore fish. The new dynamical system is represented by the following equations :

$$\begin{aligned} \frac{dC}{dt} &= \beta C(1-C) - eC - \rho CH, \\ \frac{dH}{dt} &= \alpha H\left(1-\frac{H}{K_H}\right) - \gamma LH, \\ \frac{dL}{dt} &= rL\left(1-\frac{L}{K_L}\right) - xL. \end{aligned} \tag{3}$$

The results for this adjustments are shown in Figures 6 and 7. We use the values $K_L = 3$ and $K_H = 1$ for the carrying capacities of lionfish and herbivore fish, respectively. Notice that, in contrast to the previous model, the herbivore population does not explode, rather, it settles at a value that depends on the harvesting effort x. It is also important to note that, as shown in Figure

7, harvesting lionfish at the OLY or above leads to a healthy coral (free of colonized corals) steady state.

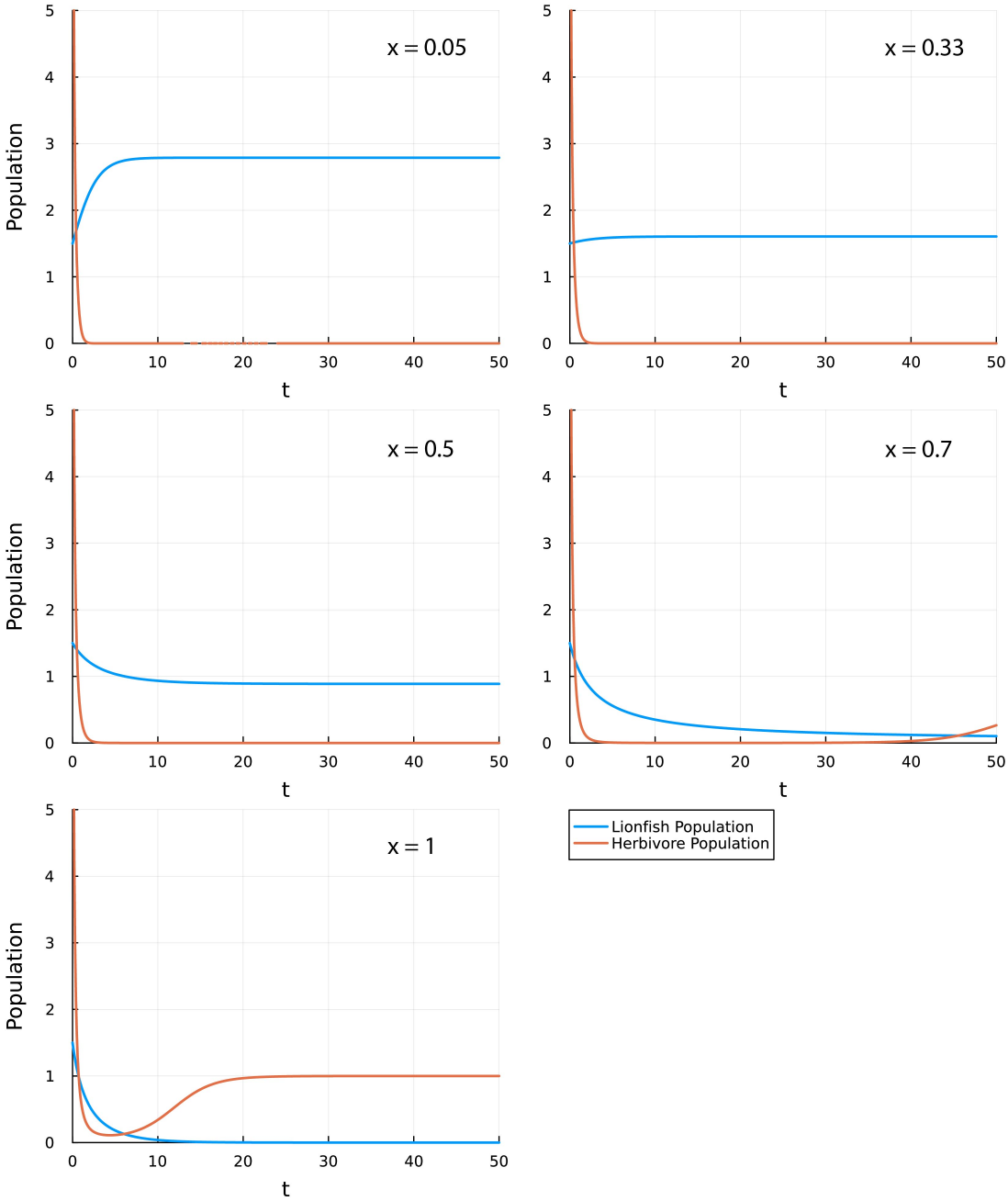

Figure 6: Evolution of lionfish and herbivore populations with logistic growth of lionfish and herbivore fish. Notice that now, neither of these populations explode, and that the behaviour depends on the harvesting effort. Furthermore, when fishing at the Optimum Lionfish Yield ($x = 0.7$), the herbivore population recovers and the health of coral reefs is restored (see Figure 7) while also maintaining the lionfish population for fishers to make profit.

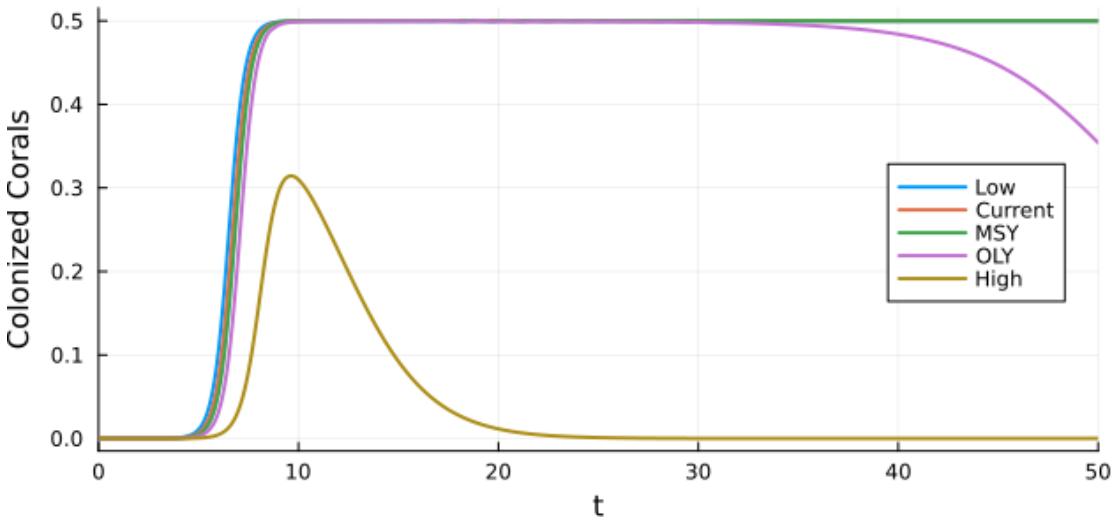

Figure 7: As before, we see the change over time of the proportion of colonized corals. Even when assuming a logistic growth (instead of linear) for herbivore fish, we see that fishing at OLY or above leads to the reduction of colonized corals at steady state.

6 Conclusion and Future Directions

In this project, we explored the cascading effects of harvesting the invasive lionfish in relation to the conservation of a coral reef ecosystem. We showed that, as expected, the harvesting rate of lionfish has major repercussions on coral reef ecosystems. To improve the health of these coral populations (uncolonized by algae), the harvesting rate of lionfish must be at the OLY or higher. Otherwise, the entire population of corals may become colonized by algae and ultimately create a hypoxic environment which is deadly to corals. We analyzed both linear and logistic growth for herbivore fish, and noticed that in either of these cases, fishing at OLY or above improves the health of coral reefs, while fishing below the OLY is detrimental to reef health. We also examined the equilibrium points of the linear-growth model. Future work should examine the equilibrium points in the logistic-growth model. Moreover, fishing at the OLY is especially favorable, given that our model demonstrates that it is able to significantly reduce the lionfish population without completely eradicating it. Fisheries can thus continue to harvest lionfish while reaping the benefits of contributing to healthy coral conservation. As corals recover, they will be able to better sustain thousands of other fish populations and protect property along the coastline.

7 Appendix

We'll now show the code used to make the simulations of the linear growth model.

```
using DifferentialEquations
using Plots

beta = 5        # algae colonization rate
e = 2.5         # extintion rate (uncoloniztion rate)
rho = 3         # rate at which hervibores feed on algae
gamma = 2       # rate at which lionfish predate on hervibores
alpha = 0.5     # growth rate of hervibores
r = 0.71        # lionfish growth rate
K = 3           # carrying capacity of lionfish

initial_colonized = 0.0001
initial_lionfish = 1.5
initial_hervibore = initial_lionfish * 10

tspan = (0.0,50.0)

# Low
x = 0.05   # harvesting rate of lionfish

function chl_low(du, u, p, t)
    C, H, L = u
    du[1] = beta * C * (1-C) - e * C - rho * C * H
    du[2] = alpha * H - gamma * L * H
    du[3] = r * L * (1 - L/K) - x * L
    nothing
end

prob_low = ODEProblem(chl_low,[initial_colonized,initial_hervibore,initial_lionfish],tspan)
sol_low = solve(prob_low, saveat = 0.1)
solC_low = sol_low[1,:]
solH_low = sol_low[2,:]
solL_low = sol_low[3,:]

# Current
x = 0.33

function chl_current(du, u, p, t)
    C, H, L = u
    du[1] = beta * C * (1-C) - e * C - rho * C * H
    du[2] = alpha * H - gamma * L * H
    du[3] = r * L * (1 - L/K) - x * L
    nothing
end

prob_current = ODEProblem(chl_current, [initial_colonized, initial_hervibore, initial_lionfish], tspan)
sol_current = solve(prob_current, saveat = 0.1)
solC_current = sol_current[1,:]
solH_current = sol_current[2,:]
solL_current = sol_current[3,:]

# MSY
x = 0.5

function chl_msy(du, u, p, t)
    C, H, L = u
    du[1] = beta * C * (1-C) - e * C - rho * C * H
    du[2] = alpha * H - gamma * L * H
    du[3] = r * L * (1 - L/K) - x * L
    nothing
end

prob_msy = ODEProblem(chl_msy, [initial_colonized, initial_hervibore, initial_lionfish], tspan)
sol_msy = solve(prob_msy, saveat = 0.1)
solC_msy = sol_msy[1,:]
solH_msy = sol_msy[2,:]
solL_msy = sol_msy[3,:]

# OLY
x = 0.7

function chl_oly(du, u, p, t)
    C, H, L = u
```

```
    du[1] = beta * C * (1-C) - e * C - rho * C * H
    du[2] = alpha * H - gamma * L * H
    du[3] = r * L * (1 - L/K) - x * L
    nothing
end

prob_oly = ODEProblem(chl_oly, [initial_colonized, initial_hervibore, initial_lionfish], tspan)
sol_oly = solve(prob_oly, saveat = 0.1)
solC_oly = sol_oly[1,:]
solH_oly = sol_oly[2,:]
solL_oly = sol_oly[3,:]

# High
x = 1

function chl_high(du, u, p, t)
    C, H, L = u
    du[1] = beta * C * (1-C) - e * C - rho * C * H
    du[2] = alpha * H - gamma * L * H
    du[3] = r * L * (1 - L/K) - x * L
    nothing
end

prob_high = ODEProblem(chl_high, [initial_colonized, initial_hervibore, initial_lionfish], tspan)
sol_high = solve(prob_high, saveat = 0.1)
solC_high = sol_high[1,:]
solH_high = sol_high[2,:]
solL_high = sol_high[3,:]

width = 900
height = 1000

l = @layout [a  b ; c  d ; e f]

p1 = plot(sol_low.t, solL_low, linewidth = 2, label = false,
        legend = :outertopleft, color = 1, ls=:solid, xlims = (0,50), size = (width, height), ylims = (0,5),
    xlabel = "t", ylabel = "Population")
plot!(sol_low.t, solH_low, linewidth = 2, label = false,
        legend = :outertopleft, color = 2, ls=:solid)

p2 = plot(sol_current.t, solL_current, linewidth = 2, label = false,
        legend = :outertopleft, color = 1, ls=:solid, xlims = (0,50), ylims = (0,5),
        xlabel = "t", ylabel = "Population")
plot!(sol_current.t, solH_current, linewidth = 2, label = false,
        legend = :outertopleft, color = 2, ls=:solid)

p3 = plot(sol_msy.t, solL_msy, linewidth = 2, label = false,
        legend = :outertopleft, color = 1, ls=:solid, xlims = (0,50), ylims = (0,5),
    xlabel = "t", ylabel = "Population")
plot!(sol_msy.t, solH_msy, linewidth = 2, label = false,
        legend = :outertopleft, color = 2, ls=:solid)

p4 = plot(sol_oly.t, solL_oly, linewidth = 2, label = false,
        legend = :outertopleft, color = 1, ls=:solid, xlims = (0,50), ylims = (0,5),
    xlabel = "t", ylabel = "Population")
plot!(sol_oly.t, solH_oly, linewidth = 2, label = false,
        legend = :outertopleft, color = 2, ls=:solid, xlims = (0,50))

p5 = plot(sol_high.t, solL_high, linewidth = 2,
        legend = :outertopleft, color = 1, ls=:solid, xlims = (0,50), ylims = (0,5),
    xlabel = "t", ylabel = "Population", label = false)
plot!(sol_high.t, solH_high, linewidth = 2,
        legend = :outertopleft, color = 2, ls=:solid, xlims = (0,50), ylims = (0,5), label = false)

p6 = plot(sol_high.t, solL_high, linewidth = 2,
        legend = :outertopleft, color = 1, ls=:solid, xlims = (0,50), ylims = (0,5),
    xlabel = "t", ylabel = "Population", label = "Lionfish Population")
plot!(sol_high.t, solH_high, linewidth = 2,
        legend = :outertopleft, color = 2, ls=:solid, xlims = (0,50), ylims = (0,5),
    label = "Herbivore Population")

hlpop = plot(p1, p2, p3, p4, p5,p6, layout = l,legend=:outerright)

width = 630
height = 300
p2 = plot(sol_low.t, solC_low, linewidth = 2, label = "Low",
        legend = :right, color = 1, ls=:solid, xlims = (0,50), size = (width, height),
```

```
        xlabel = "t", ylabel = "Colonized Corals")
plot!(sol_current.t, solC_current, linewidth = 2, label = "Current",
        legend = :right, color = 2, ls=:solid)
plot!(sol_msy.t, solC_msy, linewidth = 2, label = "MSY",
        legend = :right, color = 3, ls=:solid)
plot!(sol_oly.t, solC_oly, linewidth = 2, label = "OLY",
        legend = :right, color = 4, ls=:solid)
plot!(sol_high.t, solC_high, linewidth = 2, label = "High",
        legend = :right, color = 5, ls=:solid)
```

Finally, we show the code for the logistic growth model simulations.

```
using DifferentialEquations
using Plots

beta = 5      # algae colonization rate
e = 2.5       # extintion rate (uncoloniztion rate)
rho = 3       # rate at which hervibores feed on algae
gamma = 2     # rate at which lionfish predate on hervibores
alpha = 0.5   # growth rate of hervibores
r = 0.71      # lionfish growth rate
KL = 3        # carrying capacity of lionfish
KH = 1

initial_colonized = 0.0001
initial_lionfish = 1.5
initial_hervibore = initial_lionfish * 10

tspan = (0.0,50.0)

# Low
x = 0.05   # harvesting rate of lionfish

function chl_low(du, u, p, t)
    C, H, L = u
    du[1] = beta * C * (1-C) - e * C - rho * C * H
    du[2] = alpha * H * (1 - H/KH) - gamma * L * H
    du[3] = r * L * (1 - L/KL) - x * L
    nothing
end

prob_low = ODEProblem(chl_low, [initial_colonized, initial_hervibore, initial_lionfish], tspan)
sol_low = solve(prob_low, saveat = 0.1)
solC_low = sol_low[1,:]
solH_low = sol_low[2,:]
solL_low = sol_low[3,:]

# Current
x = 0.33

function chl_current(du, u, p, t)
    C, H, L = u
    du[1] = beta * C * (1-C) - e * C - rho * C * H
    du[2] = alpha * H * (1 - H/KH) - gamma * L * H
    du[3] = r * L * (1 - L/KL) - x * L
    nothing
end

prob_current = ODEProblem(chl_current, [initial_colonized, initial_hervibore, initial_lionfish], tspan)
sol_current = solve(prob_current, saveat = 0.1)
solC_current = sol_current[1,:]
solH_current = sol_current[2,:]
solL_current = sol_current[3,:]

# MSY
x = 0.5

function chl_msy(du, u, p, t)
    C, H, L = u
    du[1] = beta * C * (1-C) - e * C - rho * C * H
    du[2] = alpha * H * (1 - H/KH) - gamma * L * H
    du[3] = r * L * (1 - L/KL) - x * L
    nothing
end

prob_msy = ODEProblem(chl_msy, [initial_colonized, initial_hervibore, initial_lionfish], tspan)
sol_msy = solve(prob_msy, saveat = 0.1)
```

```
solC_msy = sol_msy[1,:]
solH_msy = sol_msy[2,:]
solL_msy = sol_msy[3,:]

# OLY
x = 0.7

function chl_oly(du, u, p, t)
    C, H, L = u
    du[1] = beta * C * (1-C) - e * C - rho * C * H
    du[2] = alpha * H * (1 - H/KH) - gamma * L * H
    du[3] = r * L * (1 - L/KL) - x * L
    nothing
end

prob_oly = ODEProblem(chl_oly, [initial_colonized, initial_hervibore, initial_lionfish], tspan)
sol_oly = solve(prob_oly, saveat = 0.1)
solC_oly = sol_oly[1,:]
solH_oly = sol_oly[2,:]
solL_oly = sol_oly[3,:]

# High
x = 1

function chl_high(du, u, p, t)
    C, H, L = u
    du[1] = beta * C * (1-C) - e * C - rho * C * H
    du[2] = alpha * H * (1 - H/KH) - gamma * L * H
    du[3] = r * L * (1 - L/KL) - x * L
    nothing
end

prob_high = ODEProblem(chl_high, [initial_colonized, initial_hervibore, initial_lionfish], tspan)
sol_high = solve(prob_high, saveat = 0.1)
solC_high = sol_high[1,:]
solH_high = sol_high[2,:]
solL_high = sol_high[3,:]

width = 900
height = 1000

l = @layout [a  b ; c  d ; e f]

p1 = plot(sol_low.t, solL_low, linewidth = 2, label = false,
        legend = :outertopleft, color = 1, ls=:solid, xlims = (0,50), size = (width, height), ylims = (0,5),
    xlabel = "t", ylabel = "Population")
plot!(sol_low.t, solH_low, linewidth = 2, label = false,
        legend = :outertopleft, color = 2, ls=:solid)

p2 = plot(sol_current.t, solL_current, linewidth = 2, label = false,
        legend = :outertopleft, color = 1, ls=:solid, xlims = (0,50), ylims = (0,5),
    xlabel = "t", ylabel = "Population")
plot!(sol_current.t, solH_current, linewidth = 2, label = false,
        legend = :outertopleft, color = 2, ls=:solid)

p3 = plot(sol_msy.t, solL_msy, linewidth = 2, label = false,
        legend = :outertopleft, color = 1, ls=:solid, xlims = (0,50), ylims = (0,5),
    xlabel = "t", ylabel = "Population")
plot!(sol_msy.t, solH_msy, linewidth = 2, label = false,
        legend = :outertopleft, color = 2, ls=:solid)

p4 = plot(sol_oly.t, solL_oly, linewidth = 2, label = false,
        legend = :outertopleft, color = 1, ls=:solid, xlims = (0,50), ylims = (0,5),
    xlabel = "t", ylabel = "Population")
plot!(sol_oly.t, solH_oly, linewidth = 2, label = false,
        legend = :outertopleft, color = 2, ls=:solid, xlims = (0,50))

p5 = plot(sol_high.t, solL_high, linewidth = 2,
        legend = :outertopleft, color = 1, ls=:solid, xlims = (0,50), ylims = (0,5),
    xlabel = "t", ylabel = "Population", label = false)
plot!(sol_high.t, solH_high, linewidth = 2,
        legend = :outertopleft, color = 2, ls=:solid, xlims = (0,50), ylims = (0,5), label = false)

p6 = plot(sol_high.t, solL_high, linewidth = 2,
        legend = :outertopleft, color = 1, ls=:solid, xlims = (0,50), ylims = (0,5),
    xlabel = "t", ylabel = "Population", label = "Lionfish Population")
plot!(sol_high.t, solH_high, linewidth = 2,
```

```
        legend = :outertopleft, color = 2, ls=:solid, xlims = (0,50), ylims = (0,5), label = "Herbivore Population")

hlpoplog = plot(p1, p2, p3, p4, p5,p6, layout = 1,legend=:outerright)

width = 630
height = 300
p2 = plot(sol_low.t, solC_low, linewidth = 2, label = "Low",
        legend = :right, color = 1, ls=:solid,
        xlims = (0,50), size = (width, height), xlabel = "t", ylabel = "Colonized Corals")
plot!(sol_current.t, solC_current, linewidth = 2, label = "Current",
        legend = :right, color = 2, ls=:solid)
plot!(sol_msy.t, solC_msy, linewidth = 2, label = "MSY",
        legend = :right, color = 3, ls=:solid)
plot!(sol_oly.t, solC_oly, linewidth = 2, label = "OLY",
        legend = :right, color = 4, ls=:solid)
plot!(sol_high.t, solC_high, linewidth = 2, label = "High",
        legend = :right, color = 5, ls=:solid)
```

Bibliography

[1] The importance of coral reefs: Corals tutorial. https://oceanservice.noaa.gov/education/tutorial_corals/coral07_importance.html. Accessed on April 5, 2023.

[2] Invasive lionfish. https://flowergarden.noaa.gov/education/invasivelionfish.html. Accessed: May 4, 2023.

[3] Impacts of invasive lionfish. https://www.fisheries.noaa.gov/southeast/ecosystems/impacts-invasive-lionfish, 2022. Accessed on April 4, 2023.

[4] Alex K Bogdanoff, Kyle W Shertzer, Craig A Layman, Jennifer K Chapman, Marc L Fruitema, Jennifer Solomon, Julie Sabattis, Stephanie Green, and James A Morris. Optimum lionfish yield: a non-traditional management concept for invasive lionfish (pterois spp.) fisheries. *Biological Invasions*, 23:795–810, 2021.

[5] Holden E. Harris, William F. Patterson, Robert N. M. Ahrens, Micheal S. Allen, David D. Chagaris, and Sherry L. Larkin. The bioeconomic paradox of market-based invasive species harvest: a case study of the commercial lionfish fishery. *Biological Invasions*, Feb 2023.

[6] James Morris, L Akins, Dayne Buddo, Stephanie Green, and Ricardo Lozano. Invasive lionfish: a guide to control and management. 2012.

[7] Peter J Mumby and Robert S Steneck. Coral reef management and conservation in light of rapidly evolving ecological paradigms. *Trends in ecology & evolution*, 23(10):555–563, 2008.

[8] A Ulman, FZ Ali, HE Harris, M Adel, SAAA Mabruk, M Bariche, AC Candelmo, JK Chapman, BA Cicek, KR Clements, et al. Lessons from the western atlantic lionfish invasion to inform management in the mediterranean. front. *Mar. Sci*, 9:865162, 2022.

[9] Blai Vidiella, Josep Sardanyés, and Ricard Solé. Exploiting delayed transitions to sustain semiarid ecosystems after catastrophic shifts. *Journal of The Royal Society Interface*, 15(143):20180083, 2018.